13. When hydrogen peroxide is shaken with an acidified solution of potassium dichromate in presence of ether, the ethereal layer turns blue due to the formation of

a) Cr_2O_3 b) CrO_4^{2-} c) $CrO(O_2)_2$ d) none of these

14. For decolourisation of 1 mole of acidified $KMnO_4$, the moles of H_2O_2 required is

a) $\frac{1}{2}$ b) $\frac{3}{2}$ c) $\frac{5}{2}$ d) $\frac{7}{2}$

15. Volume strength of 1.5 N H_2O_2 is

a) 1.5 b) 4.5 c) 16.8 d) 8.4

16. The hybridisation of oxygen atom is H_2O and H_2O_2 are, respectively

a) sp and sp^3 b) sp and sp c) sp and sp^2 d) sp^3 and sp^3

17. The reaction $H_3PO_2 + D_2O \rightarrow H_2DPO_2 + HDO$ indicates that hypo-phosphorus acid is

a) tribasic acid b) dibasic acid c) mono basic acid d) none of these

18. In solid ice, oxygen atom is surrounded

a) tetrahedrally by 4 hydrogen atoms

b) octahedrally by 2 oxygen and 4 hydrogen atoms

c) tetrahedrally by 2 hydrogen and 2 oxygen atoms

d) octahedrally by 6 hydrogen atoms

19. The type of H-bonding present in ortho nitro phenol and p-nitro phenol are respectively

a) inter molecular H-bonding and intra molecular H-bonding

b) intra molecular H-bonding and inter molecular H-bonding

c) intra molecular H - bonding and no H - bonding

d) intra molecular H - bonding and intra molecular H - bonding

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